HOMEWORLD RTX

0.91.3

PUBLIC BETA 3

Installation, graphics, gameplay options, map authoring, volumetric dust, mission lighting, ray-traced shadows, build notes, and known limitations.

Windows 10/11 x64 - Direct3D 12 / DXR - Non-commercial community project

REQUIRED FOR EVERY NEW EXTRACT: manually copy your legally owned HW_Music.wxd and HW_Comp.vce beside HomeworldModern.exe before first launch. Retail game data is not included.

Contents

Installation and User Guide

What this build is

Homeworld RTX is a modern Windows x64 port of the original Homeworld engine. It reads the classic game data you already own and supplies a replacement executable, modern renderer, interface assets, and authoring overlays. It does not modify your retail BIG archives.

The beta is intentionally Windows-only. Direct3D 12, DXGI, DXR, the Windows SDK, and NVIDIA Streamline are central to the current renderer. Portability can be revisited after the native renderer reaches campaign-wide parity.

Installation

Recommended clean install

1. Download the Windows x64 ZIP from GitHub Releases.
2. Create a writable folder such as `C:\Games\Homeworld RTX`.
3. Extract every file and folder into it.
4. From your legally owned Homeworld Classic data folder, manually copy

`HW_Music.wxd` and `HW_Comp.vce` into this new folder, directly beside `HomeworldModern.exe`.

5. Launch `HomeworldModern.exe`.
6. Choose the `Homeworld.exe` belonging to Homeworld 1 Classic.

> **Required beta workaround:** the current first-run verification does not reliably locate `HW_Music.wxd` and `HW_Comp.vce` through the selected retail installation. A new extract will fail verification unless both legally owned archives are manually present beside `HomeworldModern.exe`. Steam users will normally find the originals under `Homeworld1Classic\Data`.

The program checks the executable and nearby data folder. The remembered location is stored in:

```
Documents\My Games\Homeworld Modern\installation.ini
```

You can keep Homeworld RTX outside Steam. This protects the mod from Steam verification and keeps exports easy to find.

Accepted game data

For Homeworld Remastered Collection, select:

```
...\Homeworld\Homeworld1Classic\Homeworld.exe
```

Its data set normally includes:

```
Data\Homeworld.big
Data\HW_Comp.vce
Data\HW_Music.wxd
```

Copy `Data\HW_Comp.vce` and `Data\HW_Music.wxd` into the Homeworld RTX folder. Leave the originals in place and never redistribute either archive.

For an original 1999 installation, use its `Homeworld.exe` and data directory. `Update.big` is recommended for original data, but should not be added to the Remastered Collection Classic set because its updates are already folded into `Homeworld.big`.

The optional `Movies` directory enables the original pre-mission Bink videos. Retail data is never included in this project's downloads.

Updating later

Keep personal exports and diagnostics before replacing a beta:

```
RTXExports\
Documents\My Games\Homeworld Modern\RTXLighting\
Documents\My Games\Homeworld Modern\Homeworld.cfg
```

Extract a newer release over a copy or into a fresh directory. A fresh folder is easiest to diagnose because obsolete DLLs cannot remain beside the game.

First run

With no saved profile and no command-line resolution, the game adopts the primary display's active mode and starts borderless. Interface scale is derived from resolution and constrained to the window. Examples:

Output	Automatic interface scale
1280x720	1.0x
1920x1080	1.5x
2560x1440	2.0x
3440x1440	2.0x centered safe canvas
3840x2160	3.0x

The UI, mouse hit regions, fonts, cursor, modal dialogs, and manager docks share one scaling policy. The interface remains native resolution when DLSS is used.

Video settings

Display

- **Display Mode:** borderless desktop, exclusive fullscreen, or resizable window.
- **Output Resolution:** applied on the next launch where required.
- **Refresh Rate:** automatic or a supported exclusive-mode refresh.
- **Frame Limit:** presentation cap; zero is uncapped.
- **Interface Scale:** automatic resolution-derived scale with a 50-200% modifier.
- **VSync:** on, adaptive where supported, or off.

Presentation rate does not change simulation rate. AI, physics, scripts, weapons, demos, multiplayer checksums, and save behavior remain tied to the fixed gameplay clock.

Path tracing

- **Path Tracing:** turns DXR lighting on or off.
- **Light Bounces:** one to three diffuse radiance bounces.
- **Samples:** one to four new stochastic paths per pixel per frame.
- **Normal Map Strength:** 0-400% tangent-space surface relief. Dedicated ship

normal maps are used when installed; legacy textures retain an automatic luminance-derived fallback.

- **FX Emission:** 0-200% light cast by weapons, engines, beams, explosions,

glow maps, and navigation lights. Zero removes their light, not the visible FX.

Start at one sample and two or three bounces. Increase samples before bounces when you want a steadier image; increase bounces for richer indirect fill. Normal-map strength at 100% is the tuned baseline.

The temporal accumulator rejects mismatched depth, invalid previous transforms, camera cuts, newly revealed geometry, and LOD/visibility discontinuities. Fast camera movement deliberately retains less history than a still view.

Image quality

- **Off:** no post anti-aliasing.

- **Auto / Native AA (recommended):** selects DLAA on supported NVIDIA GPUs,

XeSS AA on Intel GPUs, or FidelityFX Native AA on AMD and other adapters.

- **Auto / Quality:** selects DLSS Quality, XeSS Quality, or FSR Quality from

the active adapter and installed signed runtimes.

- **FXAA:** universal, inexpensive spatial fallback with no temporal history.
- **Universal Temporal AA:** vendor-neutral native-resolution temporal

reconstruction, resolved through the packaged FidelityFX runtime.

- **DLAA / DLSS:** NVIDIA native AA plus Quality, Balanced, Performance, and

Ultra Performance full-scene modes.

- **FSR Native AA / FSR:** FidelityFX native AA plus Quality, Balanced,

Performance, and Ultra Performance modes. The Direct3D 12 runtime is cross-vendor and is also the dependable temporal fallback.

- **XeSS AA / XeSS-SR:** Intel native AA plus Quality, Balanced, Performance,

and Ultra Performance modes. Intel hardware uses the optimized path; XeSS also provides its documented shader-model fallback on supported non-Intel hardware.

- **Brightness:** final output transfer.
- **Effect Density and Effect Budget:** control background/trail/impact complexity.
- **Frame Generation:** shown as unavailable; real Streamline DLSS-G and Reflex

are not integrated in this beta.

Temporal reconstruction does not reduce UI resolution: the HUD-less world is reconstructed first and the interface is composited afterward at display resolution. Explicit unsupported modes fall back safely; Auto avoids selecting an unavailable backend. The Windows package includes the signed runtime DLLs; the selected backend still requires a compatible current driver.

Post effects

- **Chromatic Aberration:** radial RGB separation; default off.
- **Motion Blur:** uses DXR motion vectors; default off.
- **Luminance Film Grain:** response-controlled grain that protects deep blacks

and highlights; default off.

- **God Rays:** searches the mission background for the authored bright source;

ships and planet discs cannot become false suns, and world geometry occludes the radial march.

- **Bloom:** soft scene glow, 0-400%, with 100% as the visual baseline.
- **Color Banding Filter:** stable scene-only output dithering; UI is excluded.

Lighting model

The ray pass consumes the scene rather than replacing it. It uses:

- Exact visible GEO triangles and object transforms.
- Per-triangle material identity and UVs.
- LIF color, alpha, team masks, and emissive/glow texels.
- Alpha-tested visibility for cutout materials.
- Mission HDR sky radiance and detected bright-source metadata.
- Retail HSF directional, point, spot, and ambient lights.
- Active MEX navigation lights in their transformed ship positions.

- Dynamic engines, layered trails, muzzles, projectiles, harvesting beams,

hyperspace effects, and explosions.

Static geometry is cached in BLAS records. Visible instances are submitted to a triple-buffered TLAS, which is refitted when topology remains stable. Upload buffers and shader tables are reused across frames.

Classic assets do not carry modern metalness/roughness channels. The current model derives conservative material behavior and exposes global reflectivity, roughness, and generated-normal controls. Specular ray transport is not yet a full physically based reflection system.

Modern interface

The main menu, campaign, tutorial, multiplayer, options, in-game menus, and dialogs use responsive code-native layout templates. Build, Research, and Launch are right-docked managers with separate queue, quantity, cost, total, scroll, and action regions.

With ships selected, open the tactical right-click menu and choose **Ship Dossier**. The dossier shares the right dock and shows loaded ship statistics plus a concise Fleet Archive assessment.

Fleet identity editor

Fleet Identity is a unified livery and emissive editor for the local player's ships. Select one of six channels on the left, choose hue and saturation in the large color field, set value with the vertical slider, and check the live RGB/HSV/hex material key before selecting **Apply**. The picker exposes the entire HSV range with no minimum-brightness gate, so true black and near-black hulls are valid choices.

The six independently selectable channels are:

- **Hull Base:** primary player-ship pigment across the fleet.
- **Stripe / Marking:** secondary livery and team markings.
- **Engine Emission:** engine ribbons, nozzle glow, and linked DXR light.
- **Nav Lights:** navigation-light sprites and point emitters.
- **Harvest Beam:** harvesting beam, nozzle glow, and line-light emission.
- **Hyperspace:** gate, slice, field effect, and matching DXR emission.

For the four effect channels, a single switch chooses between the authored color and a custom player color. The visible raster effect and its ray-traced light use the same selection automatically; there is no separate lighting menu to synchronize. Overrides apply only to the local player's ships. Enemy and allied fleets retain their authored identities.

Capture / Build All

This single-player Fleet Control option turns exceptional captured production ships into usable factories. Enable **Capture / Build All**, salvage an enemy carrier or mothership, and the captured hull appears with your owned factories while retaining its native build roster. In other words, you can quite literally steal a carrier or mothership and build from it.

The roster belongs to the captured factory rather than being replaced by the player's ordinary ship list. Turanic and Kadeshi production hulls therefore build their own native ships. If the option is turned off later, captured hulls remain yours and already queued jobs may finish, but foreign factories are hidden from new build selection and cannot accept new construction orders until the option is re-enabled. Multiplayer retains its original owner-race rules.

Gameplay options

These additions are opt-in and saved with the user configuration:

- **Unlimited Strike-Craft Fuel:** single-player only; normal orders and docking

behavior remain available.

- **Issue Orders While Paused:** accept tactical commands while simulation time is stopped.

- **Ship Weapon Recoil:** enable physical firing impulse.
- **Cursor Scale:** 25-100% live preview.
- **Resource Multiplier:** 1x-4x deposited RU yield and collector harvest/fill

rate; authored resource values are unchanged; single-player only.

- **Super Salvagers:** 2x mobility/agility/speed/braking and 3x effective health.
- **Capture / Build All:** captured enemy carriers and exceptional production

motherships become usable factories with their native build rosters; see the dedicated section above for exact enable/disable behavior.

Simulation-changing convenience options are kept out of multiplayer behavior.

Command-line examples

```
# 1440p resizable window, uncapped, no VSync
./HomeworldModern.exe /window /width 2560 /height 1440 /noVsync /fps 0

# 4K exclusive mode at 120 Hz
./HomeworldModern.exe /exclusive /width 3840 /height 2160 /refresh 120 /fps 120 /vsync

# Cross-vendor presentation without DXR lighting
./HomeworldModern.exe /noRaytracing /fxaa

# Heavier path tracing
./HomeworldModern.exe /raytracing /pathSamples 4 /pathBounces 3 /dlssQuality
```

Frequently useful switches:

Switch	Meaning
`/window`	Resizable window
`/borderless` or `/fullscreen`	Borderless desktop mode
`/exclusive`	Exclusive fullscreen
`/width N`, `/height N`	Requested dimensions
`/refresh N`	Exclusive refresh; zero selects automatically
`/fps N`	Presentation cap; zero is uncapped
`/uiScale N`	50-200% modifier over automatic scale
`/raytracing`, `/noRaytracing`	Enable/disable DXR lighting
`/pathSamples N`	1-4 samples per pixel per frame
`/pathBounces N`	1-3 diffuse bounces
`/fxLightStrength N`	0-200% FX/emissive light contribution
`/dlaa`, `/fxaa`, `/aaOff`	AA selection
`/dlssQuality`, `/dlssBalanced`	DLSS quality modes
`/dlssPerformance`, `/dlssUltraPerformance`	DLSS performance modes
`/vsync`, `/adaptiveVsync`, `/noVsync`	Presentation synchronization
`/vanilla`	Restore the original universe update rate

The Video and Shift+F12 interfaces are preferred for normal use because they show allowed ranges and save the configuration.

Files created by the game

Location	Purpose
`Documents\My Games\Homeworld Modern\Homeworld.cfg`	Saved options
`Documents\My Games\Homeworld Modern\installation.ini`	Validated data location
`Documents\My Games\Homeworld Modern\RTXLighting\MissionNN.rtxlight`	Per-user mission lighting overrides
`RTXExports\Global\RTX-Visual-Settings.txt`	Portable global visual/shadow export
`RTXExports\Missions\MissionNN.rtxlight`	Portable mission-light export
`RTXExports\Maps\MissionNN.rtxmap`	Non-destructive map overlay export
`RTXExports\CloudPresets\`	Reusable volumetric dust presets
`%LOCALAPPDATA%\HomeworldModern\ShaderCache`	Versioned presenter shader cache

Diagnostics

Source-tree users can launch through:

```
./Windows/run-with-log.ps1 -Configuration Release -DataPath "D:\Games\HomeworldClassic\Data"
```

The script records startup stages, stdout/stderr, arguments, OS, GPU, display, exit code, and crash dump information beneath `out\logs`, then prepares a ZIP. Use `-Windowed` when testing whether a failure is specific to fullscreen.

For a binary-release report, provide your GPU, driver version, Windows version, mission, exact settings, and reproduction steps. Never attach retail data.

Next reading

- Editing Guide
- Known Limitations
- Build Guide
- Graphics Feature Status

Ship Texture Restoration and DDS Pipeline

This tool batch-processes **ship textures only**. It ignores UI, cursors, dossier art, planets, asteroids, dust clouds, and mission skies. Originals are never modified: output is written to a separate tree with the same race/ship/LOD paths.

Why it is split by texture type

- Hull/albedo textures may use Real-ESRGAN restoration.
- Team-color, emissive, opacity, roughness, metallic, AO, height, and other masks

are copied without generative processing. A model must not invent data in them.

- Dedicated tangent-space normals are derived from the vanilla hull art at its

original spatial scale, then resized and vector-renormalized. This prevents a 4x albedo from making the effective relief frequency four times finer.

- Existing alpha is extracted before AI processing and restored at target size.
- `.lif` files are left untouched. Deployment uses full-mip BC7 DDS for color,

BC5 DDS for two-channel normals, and BC4 DDS for team masks. The loader prefers DDS and retains TGA/PNG/LIF fallbacks for incomplete or user-authored sets.

The classifier is conservative. Review `ship-texture-manifest.csv` before copying generated assets into a playable tree.

Build the deployable DDS tree

After producing and reviewing the restored hull images, build the runtime pack with `tools/convert_ship_textures_to_dds.py` and the CMake-built `ship_texture_dds.exe`. Color and normal textures receive complete mip chains; mask files remain non-generative and receive their own complete mip chains.

Every color DDS may have a sibling named `<texture>_normal.dds`. The RTX mesh loader supplies that map directly to the ray-traced material. If it is absent, the renderer deliberately falls back to its legacy generated normal so custom texture packs remain usable.

One-time setup

Python 3 and Pillow are required. To install the portable, Vulkan-based Real-ESRGAN backend from its pinned official release:

```
powershell -ExecutionPolicy Bypass -File tools\setup_realesrgan.ps1
```

The download is checksum-verified and installed under the gitignored `tools/.texture-tools` directory. It works with NVIDIA, AMD, and Intel Vulkan drivers and does not require CUDA or PyTorch.

Preview the exact queue

```
powershell -ExecutionPolicy Bypass -File tools\run_ship_texture_upscale.ps1 -PlanOnly
```

Run the queue

```
powershell -ExecutionPolicy Bypass -File tools\run_ship_texture_upscale.ps1
```

Default output is `out/upscaled-ships`. Existing outputs are skipped, making the run resumable. Use `-Overwrite` only when intentionally rebuilding generated files.

For a fast deterministic test without an AI backend:

```
powershell -ExecutionPolicy Bypass -File tools\run_ship_texture_upscale.ps1 -Backend pillow
```

Use `-Source` to point at a larger legally extracted asset tree. Only top-level ship race roots (`R1`, `R2`, `P1`, `P2`, `P3`, `Traders`, or `Ships`) and their ship/LOD descendants are admitted.

Review before deployment

Inspect seams, UV islands, hull lettering, panel lines, alpha edges, team-color boundaries, engine glow coverage, and mip behavior in motion. AI restoration can create plausible-looking but incorrect plating. Commit selected results only after an in-game comparison; never replace the only copy of a source texture.

Build the indexed runtime archive

Windows release builds can memory-map the finished DDS set from one indexed `HomeworldTextures.hwt` container instead of opening thousands of individual files during mission loading:

```
python tools\pack_runtime_textures.py "<release-directory>" out\HomeworldTextures.hwt
```

Place the archive beside `HomeworldModern.exe`. The index provides sorted, case-insensitive random access; only a requested DDS byte range is copied into the existing texture upload path. Packed DDS entries are authoritative by default so release loading does not issue thousands of failed filesystem probes. Authors can set `HW_LOOSE_TEXTURE_OVERRIDES=1` while developing to restore loose DDS priority without rebuilding the archive. Asteroid and mission DDS files currently remain loose because those specialized loaders do not all use the resource-file layer.

SeedVR2 can be connected later as an expert backend, but its 3B workflow commonly needs substantially more VRAM and is not the safe unattended default for this mixed technical texture set.

Map, Lighting, Shadow, and Volumetric Editing

This guide covers the two live authoring workspaces included in the public beta:

- **Shift+F11:** map, object, and volumetric-dust editing
- **Shift+F12:** mission key lighting, local lights, global material tuning, and shadows

Both tools work as non-destructive overlays. They do not rewrite `Homeworld.big`, `Update.big`, HSF files, or other retail archives. Save frequently and retain the exported text files in version control.

Safe authoring workflow

1. Keep the release in a writable folder outside `Program Files`.
2. Start the campaign mission you want to edit.
3. Open the relevant editor.
4. Work with the simulation and KAS mission script paused when changing layout.
5. Make one class of change at a time and test at normal gameplay speed.
6. Save/export from inside the editor.
7. Copy the human-readable files from `RTXExports` into your project.
8. Reload the mission and verify the overlay from disk.

The editor uses runtime IDs to refer to original mission objects. A major change to the underlying retail mission can invalidate those references; the original archive remains untouched and is always the fallback.

Map Editor - Shift+F11

Workspace layout

The left panel has three tabs:

- **Assets:** placeable ships, derelicts, asteroids, dust, gas, and RTX volumes.

The list is built from the table of contents of the legal BIG archives plus project procedural assets.

- **Scene:** original mission objects and objects added by the editor. Modified

and deleted states are shown explicitly.

- **Volumes:** raymarched volumetric dust volumes and their material summary.

The right panel is an inspector. The top toolbar controls mission boot, pause state, viewport mode, camera preset, save/load, and transition to Lighting.

Toolbar

Control	Behavior
Mission selector	Click advances Mission 01-16; hold Shift while clicking to go backward
Load Paused	Boots the selected mission into a clean authoring state
Sim Paused / Running	Controls gameplay simulation independently
KAS Paused / Running	Controls mission scripting independently
View	Cycles Lit, Wireframe, Bounds, and Volumes
Camera	Cycles Perspective, Top, Front, and Side
Save Map	Writes the current non-destructive overlay
Load Map	Reloads the current mission and reapplies the overlay
Lighting	Closes the map tool and opens Shift+F12 lighting authoring
Done	Closes the editor and restores the previous pause state

Camera and selection

- Hold the right mouse button and move the mouse to aim the editor camera.
- Use the mouse wheel over the viewport to fly forward/backward through the

screen-center ray.

- Click an object or a projected volume marker to select it.
- Press F to focus the camera on the selection.
- Press P in Assets to place the selected asset at cursor depth.
- Press P in Scene or Volumes to move the selection to cursor depth.

The editor intentionally avoids WASD camera movement so list/inspector input does not fight gameplay keys.

Transforming objects

Selected world objects expose a direct 3D gizmo:

- Drag colored X/Y/Z arrowheads to translate.
- Drag blue axis handles to rotate.
- E chooses rotation mode for keyboard adjustments.

- `R` chooses scale mode. Scaling is limited to resources/volumes because

arbitrary ship instance scaling is unsafe for gameplay systems.

- X, Y, or Z chooses the keyboard adjustment axis.
- Arrow Up/Down changes the selected inspector row.
- Arrow Left/Right changes the selected value.
- Control applies fine adjustment; Shift applies coarse translation/size steps.

Original objects can be transformed, marked for deletion, or left unchanged. Editor additions can be duplicated with `Ctrl+Shift+D`. Deletion is represented in the overlay rather than applied to the BIG archive.

Volumetric dust

An RTX dust volume is a real D3D12 raymarched medium. Up to 128 volumes can be authored; up to 64 enabled, non-zero-density volumes are uploaded to the live renderer.

Volume fields:

Field	Purpose	Range/notes
Enabled	Includes/excludes the volume without deleting it	On/Off
Shape	Base boundary	Sphere, Box, Ellipsoid
Position X/Y/Z	World-space center	Keyboard step 100 units
Rotate X/Y/Z	Rigid volume orientation	5-degree steps; fine mode supported
Size X/Y/Z	Shape extents	50-500,000 units
Density	Medium density	0-10
Scattering	Out-of-beam light contribution	0-10
Absorption	Light removed through the medium	0-10
Anisotropy	Forward/back scattering bias	-0.95 to 0.95
Coverage	Procedural occupancy	0-2
Color R/G/B	Linear single-scattering/albedo tint	0-8 per channel
Noise Scale	World-space feature frequency	0.00001-0.1
Noise Detail	Secondary structure strength	0-2
Wake Strength	Ship disturbance response	0x off, 1x baseline, 2x maximum
Mission Key Light	Receive the mission directional source	On/Off
Local Lights	Receive point/spot/FX lighting	On/Off
Volume Shadow	Cast volumetric shadowing	On/Off
Shape Seed	Deterministic silhouette seed	`G` randomizes
Shape Variation	Boundary deformation	0-2

Tune in this order: size and transform, coverage/noise, density, scattering and absorption, color, lighting flags, then wake response. Very high density plus large overlapping volumes can obscure the scene and sharply increase raymarch cost.

Cloud preset library

The Volumes tab exposes a reusable preset dropdown:

- **Save New:** stores the selected volume's complete material, shape, seed,
transform defaults, and lighting flags as a new preset.

- **Overwrite:** replaces the selected preset from the current volume.
- **Load:** loads the preset into the selected volume or creates a new volume.
- **Copy / Paste:** duplicates full volumes between locations; `Ctrl+C` and
`Ctrl+V` provide the same operation.

Presets and their index live beneath:

```
RTXExports\CloudPresets\
```

Map overlay format and path

Save Map writes:

```
RTXExports\Maps\MissionNN.rtxmap
```

The `RTXMAP 8` text format contains:

- `MISSION` identity
- `ADD` records for placed ships/resources
- `TRANSFORM` records for moved/rotated/scaled objects
- `DELETE` records for hidden original objects
- `VOLUME DUST` records for raymarched media

The packaged `assets\Missions\MissionNN.rtxmap` is used when no writable export exists. Writable exports take precedence, which makes iteration immediate.

Do not hand-edit runtime IDs unless you understand the mission object order. Hand editing material values is possible, but the live inspector is safer and enforces limits.

Lighting Editor - Shift+F12

The lighting editor does not pause gameplay. Open it after the mission HDR sky has loaded. Use Tab or click tabs to move between **Mission Key**, **Custom Lights**, **Global Tuning**, and **Shadows**.

Rows can be adjusted by mouse wheel, `-`/`+` buttons, or Arrow Left/Right. Arrow Up/Down selects a row.

Mission Key page

Field	Meaning
Sun Azimuth	Rotates the key around world Y; 0 degrees points along -Z
Sun Elevation	Raises/lowers the key, limited to -89 to +89 degrees
Sun Intensity	Per-mission multiplier, 0-4 in 0.05 steps
Sun Red/Green/Blue	Multipliers over detected HDR source color, 0-4
Ambient Strength	HDR-derived fill strength, 0-4
Ambient Color Mode	Auto follows HDR sky mean; Manual uses RGB below
Ambient Red/Green/Blue	Manual linear ambient tint, 0-4

`P` aims the mission key using the cursor ray. `I` temporarily solos the key by muting sky ambient, environment, authored lights, and FX. `R` restores the HDR metadata baseline and reimports retail map lights. These changes are live and remain unsaved until `S` or **Save Mission**.

Custom Lights page

When the editor opens, retail HSF local lights are adopted into the editable list. System key and ambient entries remain identifiable; editable records may be Point, Spot, or Ambient.

Field	Meaning
Select Light	Cycles key, sky ambient, imported HSF lights, and added lights
Light Type	Point, Spot, or Ambient for editable records
Enabled	Temporarily bypasses a light without deleting it
Intensity	Per-light energy, also affected by Mission-Authored global gain
Red/Green/Blue	Linear light color
Range / Radius	Point/spot falloff distance in world units
Spot Cone	Inner spot half-angle
Spot Edge	Feather outside the inner cone
Y Rotation / Yaw	Rotates a spot around world Y
Position X/Y/Z	Fine world-space placement

Shortcuts:

- `N` adds a light.
- `Delete` removes the selected editable light.
- `P` places the selected point/spot at cursor depth.
- `A` aims the selected spot at the cursor.
- `S` saves the mission override and portable export.
- `L` reloads the saved override, then adopts retail map lights when needed.

Use the on-screen world marker to verify position and direction. Prefer a small number of deliberate lights; each shadow-casting source adds visibility work.

Global Tuning page

Control	Range	What it changes
Primary Sun / Shadow	0-200%	Gain after mission-local key settings
Sky Ambient / Color Wash	0-200%	Mission ambient fill
Environment Reflections	0-200%	Sky sampling and neutral path floor
Mission-Authored Lights	0-200%	HSF and live authored lights
FX / Emissive Lights	0-200%	Weapons, engines, beams, explosions, glow maps, nav lights
Surface Reflectivity	0-200%	Stable primary-surface highlights
Surface Roughness	0-100%	Highlight spread/sharpness
Normal Map Strength	0-400%	Dedicated tangent-space relief, with a legacy generated fallback
Path-Light Exposure	50-200%	Final path-light transfer only
Color Banding Filter	0-200%	Stable scene dithering before 8-bit presentation

R restores the tuned defaults: 100% light groups, 125% reflectivity, 38% roughness, 100% normal-map strength, 100% exposure, and 100% dithering. E saves options and exports all portable global visual/shader settings.

Shadows page

Control	Range	Baseline	Guidance
Shadow Strength	0-200%	100%	Global opacity of ray-traced visibility
Sun Angular Radius	0-5 degrees	0.35	Larger values broaden distant penumbrae
Sun Shadow Samples	1-16	6	Higher steadiness at higher cost
Local Shadow Samples	1-8	2	Point/spot/FX visibility rays
Local Light Radius	0-10 degrees	0.20	Area spread for local sources
Receiver Bias	0.01-2.00	0.05	Outgoing-ray offset; fights acne
Normal Bias	0-4.00	0.08	Surface-normal offset; fights self-intersection
Contact Shadow Strength	0-200%	100%	Near-field reinforcement beneath soft shadows
Contact Distance	0-10,000	1,200	Range of contact reinforcement
Max Shadow Distance	10,000-1,000,000	500,000	Visibility-ray reach

Shadow tuning order:

1. Set sun and local sample counts for your performance target.
2. Set sun/local angular radius for the desired softness.
3. Lower receiver and normal bias until acne appears, then increase slightly.
4. Tune contact strength/distance to restore grounding lost to soft shadows.
5. Shorten maximum distance only when the mission scale permits it.

Too little bias creates self-shadow acne; too much disconnects shadows from their casters (peter-panning). High sample counts multiply path-tracing cost. R resets only the Shadows page when it is active. E exports shadows with the global visual state.

Lighting save and export paths

S or **Save Mission** writes both:

```
Documents\My Games\Homeworld Modern\RTXLighting\MissionNN.rtxlight
RTXExports\Missions\MissionNN.rtxlight
```

The first is the live per-user override. The second is the portable file to commit or share. The format begins with `RTXMISSIONLIGHTING 3` and records key direction/energy, ambient state, whether authored lights replace retail map lights, and every editable local light.

E or **Export Visuals** writes:

```
RTXExports\Global\RTX-Visual-Settings.txt
```

This contains display timing, path settings, image quality, post effects, global tuning, and shadow settings. Resolution, refresh rate, AA/DLSS mode, and frame generation are deliberately excluded so an author's export does not force hardware-specific choices on another user.

Shortcut reference

Map Editor

Key	Action
`Shift+F11` or `Esc`	Close editor
Right mouse + move	Aim camera
Mouse wheel	Fly forward/back
`P`	Place selected asset or move selection to cursor depth
`F`	Focus selection
`E`	Rotation mode
`R`	Scale mode for resources/volumes
`X`, `Y`, `Z`	Select adjustment axis
Arrow Up/Down	Select inspector row
Arrow Left/Right	Adjust value
`Ctrl`	Fine adjustment modifier
`Shift`	Coarse translation/size modifier
`Delete`	Delete/hide selection through overlay
`Ctrl+Shift+D`	Duplicate selection
`G`	Randomize selected dust silhouette
`Ctrl+C`, `Ctrl+V`	Copy/paste full dust volume
`Ctrl+S`	Save map overlay
`L`	Reload mission and overlay

Lighting Editor

Key	Action
`Shift+F12` or `Esc`	Close editor and save global options
`Tab`	Cycle Mission, Lights, Tuning, Shadows
Arrow Up/Down	Select row
Arrow Left/Right	Adjust live value
`P`	Place/aim key or place selected local light
`A`	Aim selected spot light
`N`	Add local light
`Delete`	Delete selected editable local light
`I`	Solo mission key
`R`	Reset active authoring/tuning page
`S`	Save and export mission lighting
`L`	Reload saved mission lighting
`E`	Save and export global visual/shadow settings

Publishing authored work

Share only `.rtxmap`, `.rtxlight`, cloud-preset, and visual-setting text files plus art you have the right to redistribute. Never include `Homeworld.big`, `Update.big`, voice/music archives, movies, or extracted retail assets.

For a packaged project preset, keep this layout:

```
Missions\MissionNN.rtxmap
RTXLighting\MissionNN.rtxlight
CloudPresets\index.txt
CloudPresets\*.rtxdust
```

Document the required mission/data edition, gameplay-affecting object changes, and the graphics cost of high-volume/high-light scenes.

Known Limitations and Beta Expectations

Homeworld RTX 0.91.3 Beta is a working public test build, not a finished remaster. The following boundaries are intentional and should be understood before use.

Platform

- Windows 10/11 x64 only.
- The active renderer depends on Direct3D 12, DXGI, DXR, and Windows tooling.
- There are no supported Linux, macOS, Steam Deck, or ARM64 binaries.
- DXR lighting requires capable hardware. Reconstruction is no longer

NVIDIA-only: the package contains NVIDIA Streamline, AMD FidelityFX, and Intel XeSS-SR runtimes, with capability-based Auto modes.

Required retail data

- The release contains no retail BIG archives, speech, music, or movies.
- A legal Homeworld Classic installation is required.
- **New-extract blocker:** `HW_Music.wxd` and `HW_Comp.vce` must currently be

copied manually from that legal installation into the same directory as `HomeworldModern.exe`; first-run verification otherwise fails.

- Alternate language and patched data layouts are validated structurally, but

the full matrix of editions has not been campaign-tested.

- Do not combine Remastered Collection Classic data with a separate original

`Update.big`.

Renderer

- DLAA/DLSS, FidelityFX Native AA/FSR, and XeSS AA/XeSS-SR share the same

corrected full-scene color, depth, pixel-motion, jitter, reset, and native-UI contract.

- The first cross-vendor beta deliberately leaves vendor frame generation off.

Super resolution/native AA is integrated; interpolated frames and latency middleware require separate validation.

- Reactive/transparency input is accepted by the shared interface but remains

conservative until the legacy effect renderer exposes a reliable opaque-only scene boundary. Highly translucent trails, beams, and explosions are the most important AMD/Intel test cases.

- FidelityFX machine-learning acceleration is hardware-dependent. Unsupported

adapters use the runtime's compatible provider; selecting FSR does not imply that an RX 9000-only path is active.

- Native D3D12 fixed-function compatibility rasterization is experimental and

is not parity-complete for every rare legacy state.

- HDR10 and scRGB output are not implemented; output is SDR.
- DLSS Ray Reconstruction is available on compatible NVIDIA hardware and falls

back to DLSS Super Resolution when its feature/runtime contract is rejected.

- Frame Generation is disabled. The retired generic interpolation experiment

is not presented as NVIDIA DLSS-G.

- NVIDIA Reflex markers/latency integration are not implemented.

- Full specular reflection transport and authored metalness/roughness materials are not implemented. Classic materials use deterministic approximations.
- Animated/morphed BLAS updates and uncommon transparency/reactive cases may still need work.
- Multiple importance sampling and a production neural denoiser are not present.
- Very high path sample, bounce, shadow-ray, light-count, and volumetric settings can be expensive even on recent GPUs.
- The shader cache is persisted, but driver-specific D3D12 pipeline-library blobs are deliberately not reused because cross-run driver behavior was unsafe.

Visual behavior

- The renderer forces LOD0 and removes N-LIPS distance enlargement. Dense fleets may therefore cost more GPU time than the original game.
- Path-traced lighting converges progressively. Fine noise can be visible after cuts, fast movement, visibility changes, or at low sample counts.
- Fast motion intentionally reduces temporal history to avoid ghosting.
- The shipped ship normals are deterministic tangent-space maps derived from the original hull art, so they preserve the source structure but cannot add geometry detail that was never present. Textures without a sibling normal DDS use the older luminance-derived runtime fallback.
- Post effects are artistic additions and default conservatively; they may not match every mission palette.
- UI plate crops and ultrawide safe-canvas composition still require broad localization and DPI review.

Gameplay and multiplayer

- The project aims to preserve the fixed simulation contract, but this beta has not been exhaustively checksum-tested across every campaign and multiplayer scenario.
- Unlimited fuel, resource multiplier, Super Salvagers, and Capture / Build All are single-player convenience features and are excluded or constrained for multiplayer safety.
- Legacy online service code is not a supported matchmaking service. Treat LAN functionality as experimental and use identical builds/settings where required.
- Save compatibility follows the HomeworldSDL baseline; retain backups before relying on a beta for a long campaign.

Editors

- Shift+F11 and Shift+F12 are development tools and assume familiarity with mission scale, object types, and lighting tradeoffs.
- Map overlays depend on runtime object IDs. Changes to the underlying mission can invalidate transformations/deletions.
- Only 64 enabled non-zero-density volumetric dust volumes are uploaded to the live raymarcher, though up to 128 can be authored.

- Extremely large/overlapping volumes, high density, high shadow counts, and many local lights can severely reduce performance.
- The editors do not provide undo history. Save versions of exported text files.
- Mission and KAS pause states are independent; changing a live mission without pausing scripts can produce moving targets or script-side changes.
- Export directories must be writable.

Content and support

- This is a free, non-commercial community project.
- It is not affiliated with or supported by Gearbox, Relic, Sierra, NVIDIA, Steam, or the original developers/publishers.
- Do not report project bugs to the commercial game-support teams.
- Do not upload copyrighted retail data in bug reports.
- No warranty is provided. The Relic source-code agreement in `LICENSE.txt` governs the original source baseline and includes restrictive terms.

Useful report checklist

Before opening an issue, record:

1. Windows edition/build.
2. GPU model and driver version.
3. Homeworld data edition and language.
4. Mission/mode and save provenance.
5. Display mode, resolution, refresh, VSync, and frame cap.
6. Path-tracing, sample, bounce, AA/DLSS, shadow, and post-effect settings.
7. Exact steps and whether the issue reproduces after renderer reset/restart.
8. Diagnostic logs or dump, without retail assets.

Third-Party and Content Notices

Homeworld RTX combines an original Homeworld/HomeworldSDL source baseline with new project code and dynamically distributed runtime libraries. Every component retains its own license; this notice is a guide, not a replacement for those terms.

Homeworld source baseline

The original Homeworld source and modified software are governed by the Relic Entertainment source-code agreement reproduced in `LICENSE.txt`. Its grant is limited to non-commercial use/distribution and includes additional restrictions. The project and binaries are provided free of charge.

Homeworld, Relic, Sierra, Gearbox, and related names and marks belong to their respective owners. No affiliation or endorsement is claimed.

Retail data is not distributed. Users must supply their own legal installation. Do not redistribute `Homeworld.big`, `Update.big`, `HW_Comp.vce`, `HW_Music.wxd`, or retail movies through this project.

SDL2 and SDL2_net

SDL2 and SDL2_net are used for platform, input, audio, and optional networking services. The Windows release dynamically loads `SDL2.dll`. Source and current license information are available from:

- https://www.libsdl.org/
- https://github.com/libsdl-org/SDL
- https://github.com/libsdl-org/SDL_net

The exact dependency versions are resolved by the repository's `vcpkg.json` and vcpkg baseline at build time.

FFmpeg

FFmpeg libraries provide Bink movie demuxing/decoding, timing, and video pixel conversion. The release dynamically loads `avcodec`, `avformat`, `avutil`, and `swscale` DLLs. FFmpeg is generally available under LGPL 2.1 or later, but the effective license depends on build options and external libraries.

- Project: https://ffmpeg.org/
- Source: https://github.com/FFmpeg/FFmpeg
- License overview: https://github.com/FFmpeg/FFmpeg/blob/master/LICENSE.md

The distributed DLLs are generated by the vcpkg manifest build. Corresponding source recipes, build configuration, and upstream source references are available through the pinned vcpkg dependency process used by this repository. No `--enable-nonfree` configuration is requested by this project.

NVIDIA Streamline, DLSS, and NGX runtimes

NVIDIA Streamline 2.12.0 is integrated for DLAA and DLSS reconstruction. The bootstrap downloads NVIDIA's signed SDK archive from its official GitHub release, validates a pinned SHA-256 digest, and copies the required production runtimes. NVIDIA binaries are not committed to the source repository; selected redistributable DLLs are included with the Windows binary release.

- Streamline: https://github.com/NVIDIA-RTX/Streamline
- Programming guide: https://github.com/NVIDIA-RTX/Streamline/blob/main/docs/ProgrammingGuide.md
- NVIDIA RTX SDK license: included as `licenses/NVIDIA-Streamline-license.txt`
- NVIDIA NGX/DLSS license: included as `licenses/NVIDIA-DLSS-license.txt`

NVIDIA, RTX, DLSS, and related names are trademarks of NVIDIA Corporation.

AMD FidelityFX Super Resolution

AMD FidelityFX SDK 2.3.0 supplies the signed Direct3D 12 loader and upscaler runtime used for FSR Native AA and FSR Quality/Balanced/Performance modes. The same backend is the universal temporal fallback on adapters without a preferred vendor-specific reconstruction path. Bootstrap downloads the official tagged SDK source archive and verifies its pinned SHA-256 digest.

- SDK: https://github.com/GPUOpen-LibrariesAndSDKs/FidelityFX-SDK
- License: https://github.com/GPUOpen-LibrariesAndSDKs/FidelityFX-SDK/blob/main/LICENSE.txt

AMD, FidelityFX, and FSR are trademarks of Advanced Micro Devices, Inc.

Intel XeSS

Intel XeSS SDK 3.0.2 supplies XeSS-SR native AA and reconstruction quality modes. The packaged runtime includes Intel's optimized path and its documented cross-vendor shader-model fallback. Bootstrap downloads Intel's official SDK release and verifies its pinned SHA-256 digest.

- SDK: https://github.com/intel/xess
- License: https://github.com/intel/xess/blob/main/LICENSE.txt

Intel and XeSS are trademarks of Intel Corporation.

stb headers

`src/ThirdParty/stb` contains single-header libraries from the stb project. Their dual MIT/public-domain notices are embedded in the source headers.

- https://github.com/nothings/stb

DirectX Shader Compiler and Windows SDK

DXC is a build-time dependency used to compile the DXR HLSL library. Windows, Direct3D, DirectX, DXGI, and related SDK components belong to Microsoft and are used under their applicable SDK/runtime terms.

- https://github.com/microsoft/DirectXShaderCompiler

Project artwork and generated assets

Provenance notes are located beside relevant assets:

- `assets/UI/ART_PROVENANCE.md`
- `assets/Cursors/CURSOR_PROVENANCE.md`
- `assets/Missions/SKY_PROVENANCE.md`

Some interface/cursor/mission imagery is restored, transformed, or generated from lawfully supplied Homeworld material. Distribution does not grant rights to extract or reuse that material outside this non-commercial Homeworld project.

No endorsement

References to companies, products, and APIs identify compatibility only. This project is community-developed, unsupported by those companies, and provided without warranty.